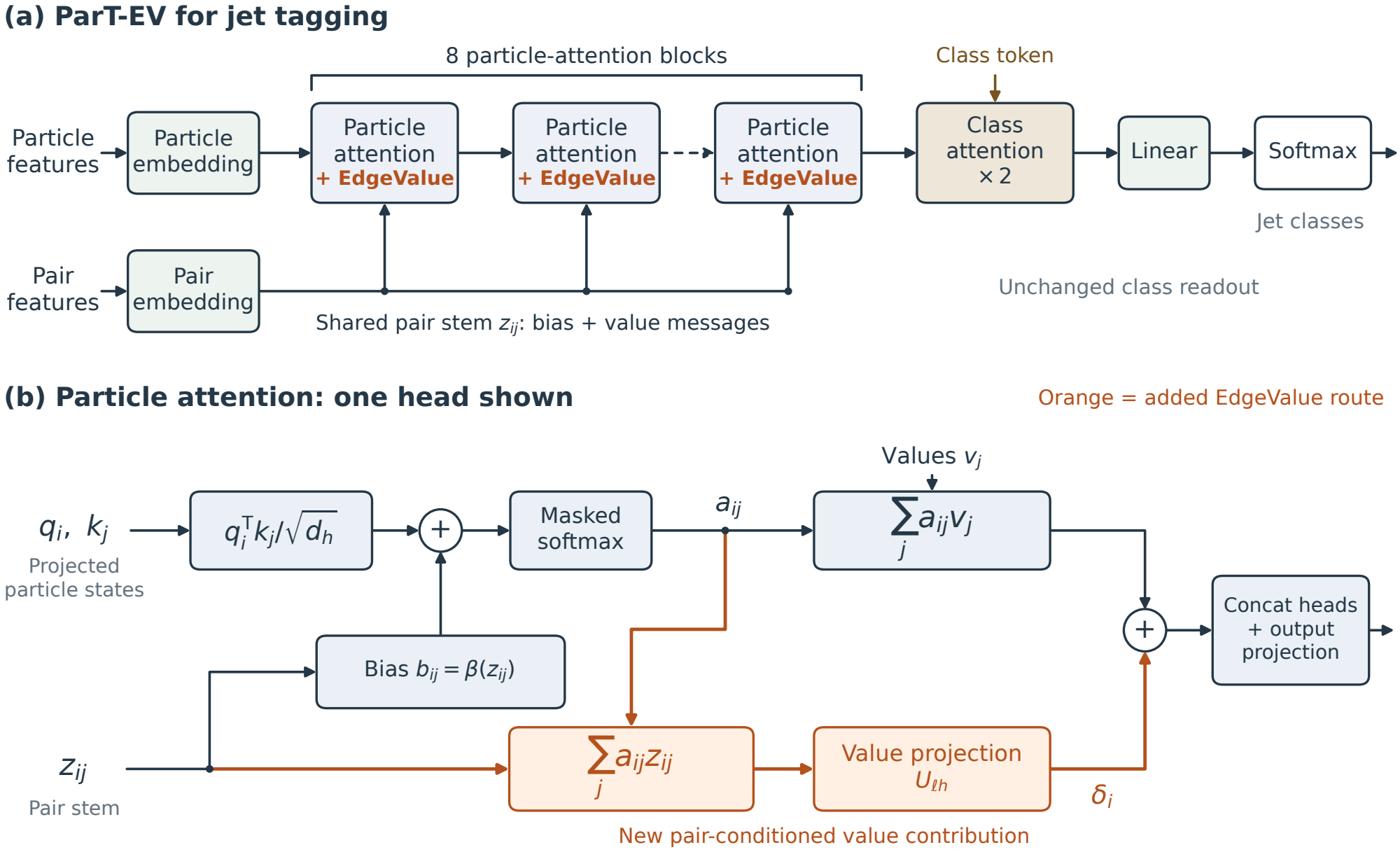
(a) ParT-EV for jet tagging
8 particle-attention blocks
Class token
Particle features
Particle embedding
Particle attention + EdgeValue
Particle attention + EdgeValue
Particle attention + EdgeValue
Class attention × 2
Linear
Softmax
Jet classes
Pair features
Pair embedding
Shared pair stem z_{ij}: bias + value messages
Unchanged class readout
(b) Particle attention: one head shown
Orange = added EdgeValue route
Values v_j
$q_i,\ k_j$
Projected particle states
$q_i^\top k_j/\sqrt{d_h}$
Masked softmax
a_{ij}
$\sum_j a_{ij} v_j$
Concat heads + output projection
Bias $b_{ij} = \beta(z_{ij})$
z_{ij}
Pair stem
$\sum_j a_{ij} z_{ij}$
Value projection $U_{\ell h}$
δ_i
New pair-conditioned value contribution